

Dr. Bharat Kumar

NUCLEAR ASTROPHYSICS

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RESEARCH PROFILE

Nuclear astrophysicist working at the interface of nuclear structure and compact-star astrophysics. My group builds relativistic mean-field energy-density functionals — the IOPB-I, G₃, NITR and NITR-I interactions — that are calibrated to finite nuclei and then pushed to the densities inside neutron stars, so that terrestrial measurements such as PREX and CREX can be confronted directly with gravitational-wave, NICER and pulsar-timing observations. Current work covers dark-matter-admixed and anisotropic compact objects, exotic core composition (hyperons, Δ baryons, quarkyonic and quark matter), radial and non-radial oscillation modes, and Bayesian and nonparametric inference of the dense-matter equation of state.

44

REFEREED ARTICLES

~1700

CITATIONS

20

H-INDEX

4

DOCTORAL STUDENTS

Rs. 21.1 L

SERB GRANT, AS PI

Citation metrics from INSPIRE-HEP, September 2026 (*i*10-index 30); Google Scholar figures are higher. Full record: 44 refereed journal articles, 1 book chapter, 22 conference proceedings, 4 preprints under review.

ACADEMIC APPOINTMENTS

2020 – Present	Assistant Professor , Department of Physics and Astronomy National Institute of Technology Rourkela, Odisha, India
2022 – 2025	Associate , Inter-University Centre for Astronomy and Astrophysics IUCAA, Pune, India
2019 – 2020	Postdoctoral Researcher , Center for Computational Sciences University of Tsukuba, Japan • Host: Prof. T. Nakatsukasa
2018 – 2019	Postdoctoral Researcher , LIGO-India Inter-University Centre for Astronomy and Astrophysics, Pune, India • Host: Prof. Sukanta Bose

EDUCATION

2014 – 2018	Ph.D. in Physics — Institute of Physics, Bhubaneswar (Homi Bhabha National Institute), India Thesis: <i>Implications of nuclear interaction for nuclear structure and astrophysics within the relativistic mean-field model</i> arXiv:1812.03306 Supervisor: Prof. S. K. Patra
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2013 – 2014	Post-M.Sc. Associateship — Institute of Physics, Bhubaneswar, India Project: <i>Shape coexistence and parity doublet in Zr isotopes</i> • Supervisor: Prof. S. K. Patra
2010 – 2012	M.Sc. in Physics — Aligarh Muslim University, Aligarh, India
2007 – 2010	B.Sc. (Hons.) in Physics — Aligarh Muslim University, Aligarh, India

RESEARCH GRANTS

2022 – 2025 Rs. 21,12,828	Physics of Neutron Stars: Constraints from Finite Nuclei to Gravitational Waves Science and Engineering Research Board (SERB), Government of India — Core Research Grant CRG/2021/000101 Role: Principal Investigator • Co-PI: Prof. B. K. Agrawal (SINP, Kolkata)
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RESEARCH INTERESTS

Nuclear Structure	Ground-state properties of finite nuclei; nuclear energy-density functionals; relativistic mean-field and Skyrme Hartree–Fock models; development of effective interactions (IOPB-I, G ₃ , NITR, NITR-I); parity doublets and shape coexistence; multifragmentation and fission yields; density dependence of the symmetry energy; neutron skins and the PREX/CREX constraints; equation of state of dense and warm nuclear matter.
Compact-Star Astrophysics	Static, rotating and anisotropic neutron-star structure — mass, radius, moment of inertia, quadrupole deformation, tidal deformability; radial and non-radial (f -, p -, g -mode) oscillations; proto-neutron stars at finite entropy; exotic core composition (hyperons, Δ baryons, quarkyonic and quark matter); dark-matter-admixed compact objects and two-fluid models; spacetime curvature and compact stars in modified gravity; equation-of-state-insensitive (universal) relations.
Computational Methods	Bayesian inference and Markov-chain Monte Carlo; nonparametric equation-of-state representations; machine-learning-assisted parameter estimation; large-scale numerical solution of the TOV and stellar-perturbation equations.

RESEARCH HIGHLIGHTS

Five papers that best represent the contribution; the complete list follows. ★ denotes lead or corresponding authorship.

- GW170817: constraining the nuclear matter equation of state from the neutron star tidal deformability** 298 citations
 T. Malik, N. Alam, M. Fortin, C. Providência, B. K. Agrawal, T. K. Jha, **Bharat Kumar**, S. K. Patra
Phys. Rev. C 98, 035804 (2018) [10.1103/PhysRevC.98.035804](https://doi.org/10.1103/PhysRevC.98.035804)
 Established the correlations that turn a measured tidal deformability into bounds on nuclear incompressibility and symmetry energy — one of the most cited routes from GW170817 to laboratory nuclear physics.
- ★ New relativistic effective interaction for finite nuclei, infinite nuclear matter and neutron stars** 153 citations
Bharat Kumar, B. K. Agrawal, S. K. Patra
Phys. Rev. C 97, 045806 (2018) [10.1103/PhysRevC.97.045806](https://doi.org/10.1103/PhysRevC.97.045806)
 Introduced the IOPB-I interaction, now used internationally as a standard relativistic mean-field parameterisation spanning finite nuclei to neutron-star matter.
- Effects of dark matter on the nuclear and neutron star matter** 116 citations

H. C. Das, A. Kumar, **Bharat Kumar**, S. K. Biswal, T. Nakatsukasa, A. Li, S. K. Patra

Mon. Not. R. Astron. Soc. **495**, 4893 (2020) [10.1093/mnras/staa1435](https://doi.org/10.1093/mnras/staa1435)

First systematic treatment of WIMP dark matter inside the relativistic mean-field description of dense matter; opened the group's ongoing programme on dark-matter-admixed compact objects.

4. ★ **Inferring neutron star properties from GW170817 with universal relations** 110 citations

Bharat Kumar, P. Landry

Phys. Rev. D **99**, 123026 (2019) [10.1103/PhysRevD.99.123026](https://doi.org/10.1103/PhysRevD.99.123026)

Used equation-of-state-insensitive relations to extract radii, moments of inertia and binding energies of the GW170817 components without assuming a specific equation of state.

5. ★ **Constraints on the moment of inertia of PSR J0737–3039A from GW170817** 93 citations

P. Landry, **Bharat Kumar**

Astrophys. J. Lett. **868**, L22 (2018) [10.3847/2041-8213/aabee76](https://doi.org/10.3847/2041-8213/aabee76)

Predicted the moment of inertia of the double-pulsar primary ahead of its measurement, linking gravitational-wave and radio-timing observables.

PUBLICATIONS

Reverse chronological. Own name **Bharat Kumar**; ★ marks lead or corresponding authorship. Every entry carries a resolvable DOI and, where available, the arXiv identifier. Bibliographic data verified against INSPIRE-HEP, September 2026.

Refereed Journal Articles

44 papers

- Spacetime curvature as a probe of exotic core phases in neutron stars within modified gravity
S. Ghosh, **Bharat Kumar**, S. Mahapatra
Phys. Rev. D **113**, 024070 (2026) [10.1103/rjzl-pcr4](https://doi.org/10.1103/rjzl-pcr4) | [arXiv:2508.08866](https://arxiv.org/abs/2508.08866) IF 5.30
- Astrophysical constraints on neutron star f -modes with a nonparametric equation of state representation
S. R. Mohanty, U. Mali, H. C. Das, **Bharat Kumar**, P. Landry
Phys. Rev. D **112**, 123043 (2025) [10.1103/5v5y-d3v8](https://doi.org/10.1103/5v5y-d3v8) | [arXiv:2410.16689](https://arxiv.org/abs/2410.16689) IF 5.30
- Characterizing the nuclear models informed by PREX and CREX: a view from Bayesian inference
T. Zhao, Z. Lin, **Bharat Kumar**, A. W. Steiner, M. Prakash
Phys. Rev. Research **7**, 043335 (2025) [10.1103/472x-9cxj](https://doi.org/10.1103/472x-9cxj) | [arXiv:2406.05267](https://arxiv.org/abs/2406.05267) IF 4.20
- The role of adiabatic sound speeds in neutron star radial oscillations and stability
S. Ghosh, T. Zhao, **Bharat Kumar**, S. R. Mohanty
JCAP **09**, 025 (2025) [10.1088/1475-7516/2025/09/025](https://doi.org/10.1088/1475-7516/2025/09/025) | [arXiv:2401.08347](https://arxiv.org/abs/2401.08347) IF 7.28
- Effects of asymmetric dark matter on a magnetized neutron star: a two-fluid approach
P. Routaray, V. Parmar, H. C. Das, **Bharat Kumar**, G. F. Burgio, H.-J. Schulze
Phys. Rev. D **111**, 103045 (2025) [10.1103/PhysRevD.111.103045](https://doi.org/10.1103/PhysRevD.111.103045) | [arXiv:2412.21097](https://arxiv.org/abs/2412.21097) IF 5.30
- Decoding dark matter admixed neutron stars: from static structure to rotational deformation
P. Routaray, A. Chakrawarty, N. K. Patra, **Bharat Kumar**
Phys. Dark Univ. **50**, 102093 (2025) [10.1016/j.dark.2025.102093](https://doi.org/10.1016/j.dark.2025.102093) | [arXiv:2409.02131](https://arxiv.org/abs/2409.02131) IF 6.40
- Correlation between the curvature and some properties of the neutron star
S. K. Biswal, H. C. Das, A. Kumar, **Bharat Kumar**, R. Jena, P. Dash, S. K. Patra
Int. J. Mod. Phys. E **34**, 2550009 (2025) [10.1142/S0218301325500090](https://doi.org/10.1142/S0218301325500090) | [arXiv:2012.13673](https://arxiv.org/abs/2012.13673) IF 1.17
- Exploring the impact of the σ -cut scheme on neutron star oscillations and thermal relaxations
P. J. Kalita, P. Routaray, K. Singha, **Bharat Kumar**, B. K. Agrawal
Phys. Rev. C **110**, 045807 (2024) [10.1103/PhysRevC.110.045807](https://doi.org/10.1103/PhysRevC.110.045807) IF 3.40

9. Constraining neutron star properties and dark matter admixture with the NITR-I equation of state
P. Routaray, H. C. Das, J. A. Pattnaik, **Bharat Kumar**
Int. J. Mod. Phys. E 33, 2450052 (2024) [DOI](#) 10.1142/S0218301324500526 | [arXiv:2307.12748](#) IF 1.17
10. Exploring the macroscopic properties of proto-neutron stars: effects of entropy and lepton fraction
S. Ghosh, S. Shaikh, P. J. Kalita, P. Routaray, **Bharat Kumar**, B. K. Agrawal
Nucl. Phys. B 1008, 116697 (2024) [DOI](#) 10.1016/j.nuclphysb.2024.116697 | [arXiv:2307.06892](#) IF 3.04
11. Unstable anisotropic neutron stars: probing the limits of gravitational collapse
S. R. Mohanty, S. Ghosh, **Bharat Kumar**
Phys. Rev. D 109, 123039 (2024) [DOI](#) 10.1103/PhysRevD.109.123039 | [arXiv:2304.02439](#) IF 5.30
12. Probing the impact of Δ baryons on nuclear matter and non-radial oscillations in neutron stars
P. J. Kalita, P. Routaray, S. Ghosh, **Bharat Kumar**, B. K. Agrawal
JCAP 04, 065 (2024) [DOI](#) 10.1088/1475-7516/2024/04/065 | [arXiv:2308.09008](#) IF 7.28
13. The impact of anisotropy on neutron star properties: insights from I - f - C universal relations [Invited]
S. R. Mohanty, S. Ghosh, P. Routaray, H. C. Das, **Bharat Kumar**
JCAP 03, 054 (2024) [DOI](#) 10.1088/1475-7516/2024/03/054 | [arXiv:2305.15724](#) IF 7.28
14. Investigating dark matter-admixed neutron stars with the NITR equation of state in light of PSR J0952–0607
P. Routaray, S. R. Mohanty, H. C. Das, S. Ghosh, P. J. Kalita, V. Parmar, **Bharat Kumar**
JCAP 10, 073 (2023) [DOI](#) 10.1088/1475-7516/2023/10/073 | [arXiv:2304.05100](#) IF 7.28
15. Probing the impact of WIMP dark matter on universal relations, GW170817 posterior, and radial oscillations
P. Routaray, A. Quddus, K. Chakravarti, **Bharat Kumar**
Mon. Not. R. Astron. Soc. 525, 5492 (2023) [DOI](#) 10.1093/mnras/stad2628 | [arXiv:2202.04364](#) IF 5.23
16. Radial oscillations of dark matter admixed neutron stars
P. Routaray, H. C. Das, S. Sen, **Bharat Kumar**, G. Panotopoulos, T. Zhao
Phys. Rev. D 107, 103039 (2023) [DOI](#) 10.1103/PhysRevD.107.103039 | [arXiv:2211.12808](#) IF 5.30
17. Radial oscillations in neutron stars from unified hadronic and quarkyonic equations of state [Invited]
S. Sen, S. Kumar, A. Kunjipurayil, P. Routaray, S. Ghosh, P. J. Kalita, T. Zhao, **Bharat Kumar**
Galaxies 11, 60 (2023) [DOI](#) 10.3390/galaxies11020060 | [arXiv:2205.02076](#) IF 3.30
18. Theoretical studies on structural properties and decay modes of $^{284-375}_{119}$ isotopes
A. A. Rather, M. Ikram, I. A. Rather, M. Imran, A. A. Usmani, **Bharat Kumar**, K. P. Santhosh, S. K. Patra
Eur. Phys. J. Plus 138, 467 (2023) [DOI](#) 10.1140/epjp/s13360-023-03959-6 | [arXiv:1709.07311](#) IF 3.75
19. Impact of the equation of state on f - and p -mode oscillations of neutron stars
A. Kunjipurayil, T. Zhao, **Bharat Kumar**, B. K. Agrawal, M. Prakash
Phys. Rev. D 106, 063005 (2022) [DOI](#) 10.1103/PhysRevD.106.063005 | [arXiv:2205.02081](#) IF 5.30
20. Dark matter effects on the compact star properties [Invited Review]
H. C. Das, A. Kumar, **Bharat Kumar**, S. K. Patra
Galaxies 10, 14 (2022) [DOI](#) 10.3390/galaxies10010014 | [arXiv:2112.14198](#) IF 3.17
21. Universal relations between the quasinormal modes of neutron stars and tidal deformability
H. Sotani, **Bharat Kumar**
Phys. Rev. D 104, 123002 (2021) [DOI](#) 10.1103/PhysRevD.104.123002 | [arXiv:2109.08145](#) IF 5.30
22. BigApple force and its implications to finite nuclei and astrophysical objects
H. C. Das, A. Kumar, **Bharat Kumar**, S. K. Biswal, S. K. Patra
Int. J. Mod. Phys. E 30, 2150088 (2021) [DOI](#) 10.1142/S0218301321500889 | [arXiv:2009.10690](#) IF 1.17
23. Impacts of dark matter on the curvature of the neutron star
H. C. Das, A. Kumar, **Bharat Kumar**, S. K. Biswal, S. K. Patra
JCAP 01, 007 (2021) [DOI](#) 10.1088/1475-7516/2021/01/007 | [arXiv:2007.05382](#) IF 7.28
24. Warm dense matter and cooling of supernovae remnants

A. Kumar, H. C. Das, S. K. Biswal, **Bharat Kumar**, S. K. Patra

Eur. Phys. J. C 80, 775 (2020) [DOI](#) 10.1140/epjc/s10052-020-8353-4 | [arXiv:2005.08320](#) IF 4.39

25. ★ GW170817 constraints on the properties of a neutron star in the presence of WIMP dark matter
A. Quddus, G. Panotopoulos, **Bharat Kumar**, S. Ahmad, S. K. Patra
J. Phys. G 47, 095202 (2020) [DOI](#) 10.1088/1361-6471/ab9d36 | [arXiv:1902.00929](#) IF 2.41
26. Effects of dark matter on the nuclear and neutron star matter
H. C. Das, A. Kumar, **Bharat Kumar**, S. K. Biswal, T. Nakatsukasa, A. Li, S. K. Patra
Mon. Not. R. Astron. Soc. 495, 4893 (2020) [DOI](#) 10.1093/mnras/staa1435 | [arXiv:2002.00594](#) IF 5.54
27. ★ Inferring neutron star properties from GW170817 with universal relations
Bharat Kumar, P. Landry
Phys. Rev. D 99, 123026 (2019) [DOI](#) 10.1103/PhysRevD.99.123026 | [arXiv:1902.04557](#) IF 5.30
28. ★ Constraints on the moment of inertia of PSR J0737–3039A from GW170817
P. Landry, **Bharat Kumar**
Astrophys. J. Lett. 868, L22 (2018) [DOI](#) 10.3847/2041-8213/aee76 | [arXiv:1807.04727](#) IF 11.70
29. GW170817: constraining the nuclear matter equation of state from the neutron star tidal deformability
T. Malik, N. Alam, M. Fortin, C. Providência, B. K. Agrawal, T. K. Jha, **Bharat Kumar**, S. K. Patra
Phys. Rev. C 98, 035804 (2018) [DOI](#) 10.1103/PhysRevC.98.035804 | [arXiv:1805.11963](#) IF 3.40
30. Decay properties and reaction dynamics of zirconium isotopes in the relativistic mean-field model
M. Panigrahi, R. N. Panda, **Bharat Kumar**, S. K. Patra
Int. J. Mod. Phys. E 27, 1850012 (2018) [DOI](#) 10.1142/S021830131850012X IF 1.17
31. ★ New relativistic effective interaction for finite nuclei, infinite nuclear matter and neutron stars
Bharat Kumar, B. K. Agrawal, S. K. Patra
Phys. Rev. C 97, 045806 (2018) [DOI](#) 10.1103/PhysRevC.97.045806 | [arXiv:1711.04940](#) IF 3.40
32. A study of multi- Λ hypernuclei within the spherical relativistic mean-field approach
A. A. Rather, M. Ikram, A. A. Usmani, **Bharat Kumar**, S. K. Patra
Braz. J. Phys. 47, 628 (2017) [DOI](#) 10.1007/s13538-017-0525-9 | [arXiv:1708.00409](#) IF 1.40
33. ★ New parameterization of the effective field theory motivated relativistic mean field model
Bharat Kumar, S. K. Singh, B. K. Agrawal, S. K. Patra
Nucl. Phys. A 966, 197 (2017) [DOI](#) 10.1016/j.nuclphysa.2017.07.001 | [arXiv:1705.02621](#) IF 1.70
34. ★ Relative mass distributions of neutron-rich thermally fissile nuclei within a statistical model
Bharat Kumar, M. T. Senthil Kannan, M. Balasubramaniam, B. K. Agrawal, S. K. Patra
Phys. Rev. C 96, 034623 (2017) [DOI](#) 10.1103/PhysRevC.96.034623 | [arXiv:1704.08105](#) IF 3.40
35. Relative fragmentation in ternary systems within the temperature-dependent relativistic mean-field approach
M. T. Senthil Kannan, **Bharat Kumar**, M. Balasubramaniam, B. K. Agrawal, S. K. Patra
Phys. Rev. C 95, 064613 (2017) [DOI](#) 10.1103/PhysRevC.95.064613 | [arXiv:1612.01667](#) IF 3.40
36. ★ Tidal deformability of neutron and hyperon stars within relativistic mean field equations of state
Bharat Kumar, S. K. Biswal, S. K. Patra
Phys. Rev. C 95, 015801 (2017) [DOI](#) 10.1103/PhysRevC.95.015801 | [arXiv:1609.08863](#) IF 3.40
37. Structural and decay properties of $Z = 132, 138$ superheavy nuclei
A. A. Rather, M. Ikram, A. A. Usmani, **Bharat Kumar**, S. K. Patra
Eur. Phys. J. A 52, 372 (2016) [DOI](#) 10.1140/epja/i2016-16372-x | [arXiv:1611.00232](#) IF 2.80
38. Quest for magicity in hypernuclei
M. Ikram, A. A. Rather, **Bharat Kumar**, S. K. Biswal, S. K. Patra
Int. J. Mod. Phys. E 25, 1650103 (2016) [DOI](#) 10.1142/S0218301316501032 | [arXiv:1607.04865](#) IF 1.17
39. Effects of the isovector scalar meson on neutron stars with and without hyperons
S. K. Biswal, **Bharat Kumar**, S. K. Patra

Int. J. Mod. Phys. E 25, 1650090 (2016) [DOI](#) 10.1142/S0218301316500907 | [arXiv:1602.08888](#) IF 1.17

40. Nuclear structure and decay properties of even–even nuclei in the $Z = 70$ –80 drip-line region

S. Mahapatro, C. Lahiri, **Bharat Kumar**, R. N. Mishra, S. K. Patra

Int. J. Mod. Phys. E 25, 1650062 (2016) [DOI](#) 10.1142/S0218301316500622 | [arXiv:1512.04665](#) IF 1.17

41. ★ Modes of decay in neutron-rich nuclei

Bharat Kumar, S. K. Biswal, S. K. Singh, C. Lahiri, S. K. Patra

Int. J. Mod. Phys. E 25, 1650020 (2016) [DOI](#) 10.1142/S0218301316500208 IF 1.17

42. ★ Examining the stability of thermally fissile Th and U isotopes

Bharat Kumar, S. K. Biswal, S. K. Singh, S. K. Patra

Phys. Rev. C 92, 054314 (2015) [DOI](#) 10.1103/PhysRevC.92.054314 | [arXiv:1508.00391](#) IF 3.40

43. Properties of superheavy nuclei with $Z = 124$

M. S. Mehta, H. Kaur, **Bharat Kumar**, S. K. Patra

Phys. Rev. C 92, 054305 (2015) [DOI](#) 10.1103/PhysRevC.92.054305 | [arXiv:1510.08312](#) IF 3.40

44. ★ Shape coexistence and parity doublet in Zr isotopes

Bharat Kumar, S. K. Singh, S. K. Patra

Int. J. Mod. Phys. E 24, 1550017 (2015) [DOI](#) 10.1142/S0218301315500172 | [arXiv:1409.4645](#) IF 1.17

Book Chapters

1. Recent parametrizations in the relativistic mean-field formalism

K. C. Naik, R. N. Panda, **Bharat Kumar**, S. K. Patra

Nuclear Structure Physics, CRC Press (2020) ISBN 978-0-367-25610-4

Preprints and Manuscripts Under Review

1. Trace anomaly and interior curvature of neutron stars in energy-momentum-squared gravity

R. Swain, S. Ghosh, **Bharat Kumar**

[arXiv:2606.20203](#)

2. Observable signatures of a quarkyonic phase in neutron stars

P. J. Kalita, T. Malik, T. Zhao, **Bharat Kumar**, J. M. Lattimer

[arXiv:2510.23405](#)

3. Neutron star properties in $f(\mathcal{R})$ gravity

P. Roy, **Bharat Kumar**

[arXiv:2306.06759](#)

4. ★ Structure effects on fission yields

Bharat Kumar, M. T. Senthil Kannan, M. Balasubramaniam, B. K. Agrawal, S. K. Patra

[arXiv:1701.00731](#)

CONFERENCE AND SYMPOSIUM PROCEEDINGS

1. Neutron star radial stability: insights from sound speed

S. Ghosh, S. R. Mohanty, T. Zhao, **Bharat Kumar**

Springer Proc. Phys. 432, 99 (2025) [DOI](#) 10.1007/978-981-95-1513-4_20

2. Universal relation in dark matter admixed neutron stars

P. Routaray, H. C. Das, **Bharat Kumar**

Proc. DAE Symp. Nucl. Phys. 67, 815 (2024)

3. A study on the σ - Δ coupling strength's impact on tidal deformability and f -mode oscillations of neutron stars
P. J. Kalita, **Bharat Kumar**
Proc. DAE Symp. Nucl. Phys. 67, 821 (2024)
4. Influence of entropy on the f -mode oscillation of proto-neutron stars
S. Ghosh, **Bharat Kumar**, B. K. Agrawal
Proc. DAE Symp. Nucl. Phys. 67, 781 (2024)
5. Radial oscillations of the dark matter admixed neutron star
P. Routaray, S. Sen, **Bharat Kumar**, H. C. Das
Proc. DAE Symp. Nucl. Phys. 66, 794 (2023)
6. The f -mode and the density dependence of the symmetry energy
A. Kunjipurayil, T. Zhao, **Bharat Kumar**, B. K. Agrawal, M. Prakash
Proc. DAE Symp. Nucl. Phys. 66, 728 (2023)
7. ★ Neutron skins of heavy nuclei and tidal deformability of neutron stars
Bharat Kumar
JPS Conf. Proc. 31, 011052 (2020) [10.7566/JPSCP.31.011052](https://doi.org/10.7566/JPSCP.31.011052) | [arXiv:1908.02909](https://arxiv.org/abs/1908.02909)
8. GW170817 constraints on equations of state of a neutron star in the presence of WIMP dark matter
A. Qudus, G. Panotopoulos, **Bharat Kumar**, S. Ahmad, S. K. Patra
JPS Conf. Proc. 31, 011053 (2020) [10.7566/JPSCP.31.011053](https://doi.org/10.7566/JPSCP.31.011053)
9. Properties of a neutron star in the presence of dark matter
A. Qudus, G. Panotopoulos, **Bharat Kumar**, S. Ahmad, S. K. Patra
Proc. DAE Symp. Nucl. Phys. 63, 810 (2018)
10. Determination of the nuclear incompressibility and symmetry energy from neutron star tides
T. Malik, N. Alam, M. Fortin, C. Providência, B. K. Agrawal, T. K. Jha, **Bharat Kumar**, S. K. Patra
Proc. DAE Symp. Nucl. Phys. 63, 816 (2018)
11. ★ Effective relativistic mean field model for finite nuclei and neutron stars
Bharat Kumar, B. K. Agrawal, S. K. Patra
Proc. DAE Symp. Nucl. Phys. 62, 712 (2017) [arXiv:1709.01252](https://arxiv.org/abs/1709.01252)
12. ★ Tidal deformability of neutron and hyperon stars
Bharat Kumar, S. K. Patra
Proc. DAE Symp. Nucl. Phys. 62, 21 (2017)
13. ★ Curvature of a neutron star
Bharat Kumar, S. K. Biswal, S. K. Patra
Proc. DAE-BRNS Symp. Nucl. Phys. 61, 916 (2016)
14. ★ Tidal effects in equal-mass binary neutron stars
Bharat Kumar, S. K. Biswal, S. K. Patra
Proc. DAE-BRNS Symp. Nucl. Phys. 61, 868 (2016)
15. ★ Evolution of the $N = 32, 34$ shell closure in relativistic mean field theory
Bharat Kumar, S. K. Biswal, S. K. Patra
Proc. DAE-BRNS Symp. Nucl. Phys. 61, 196 (2016)
16. Competition between α , β decay and spontaneous fission in $Z = 132$ superheavy nuclei
A. A. Rather, M. Ikram, **Bharat Kumar**, S. K. Biswal, S. K. Patra
Proc. DAE-BRNS Symp. Nucl. Phys. 61, 202 (2016)
17. Search for Λ shell closures in multi- Λ hypernuclei
A. A. Rather, M. Ikram, M. Imran, **Bharat Kumar**, S. K. Biswal, S. K. Patra

Proc. DAE-BRNS Symp. Nucl. Phys. 61, 178 (2016)

18. Effects of the δ meson on the maximum mass of the hyperon star
S. K. Biswal, **Bharat Kumar**, S. K. Patra
Proc. DAE-BRNS Symp. Nucl. Phys. 61, 912 (2016)
19. ★ I -Love relation in low-mass neutron stars
Bharat Kumar, S. K. Biswal, S. K. Patra
NCNAP-2016, Central University of Jharkhand, Ranchi, India, 4–6 October 2016
20. ★ β -decay half-life of Th and U isotopes
Bharat Kumar, S. K. Biswal, S. K. Singh, S. K. Patra
Proc. DAE Symp. Nucl. Phys. 60, 406 (2015)
21. ★ Nucleus and its application: neutron-rich thermally fissile nuclei
Bharat Kumar, S. K. Biswal, S. K. Patra
National Seminar on Recent Trends in Physics and 33rd Convention of the Orissa Physical Society (2015)
22. ★ Analysis of parity doublet in medium mass nuclei
Bharat Kumar, S. K. Singh, S. K. Patra
Proc. DAE Symp. Nucl. Phys. 59, 96 (2014)

RESEARCH SUPERVISION

Doctoral Students — Completed

- | | |
|-------------|---|
| 2021 – 2026 | Dr. Pinku Routaray — Dark matter effects in neutron stars: from static structure to rotational deformation
Thesis defended July 2026. Joining the Kavli Institute for Astronomy and Astrophysics (KIAA), Peking University, China, as a Postdoctoral Fellow.
★ Best Poster Award, DAE-BRNS High Energy Physics Symposium, BHU, December 2024 |
|-------------|---|

Doctoral Students — In Progress

- | | |
|-----------------|---|
| Since 2022 | Sayantana Ghosh — Neutron-star oscillations, radial stability and spacetime curvature
M.Sc., Banaras Hindu University |
| Since 2022 | Probit Jyoti Kalita — Exotic degrees of freedom and non-radial oscillations in neutron stars
M.Sc., Tezpur University |
| Since July 2026 | Subha Mitra — Nuclear astrophysics and compact-star structure |

Masters and Undergraduate Projects

- | | |
|-------------------------------|--|
| Sailesh Ranjan Mohanty | M.Sc. Impacts of anisotropy on neutron-star properties: insights from non-adiabatic gravitational collapse |
| Athul K. P. | M.Sc. Impact of the equation of state on f - and p -mode oscillations of neutron stars
Current position: Ph.D., Florida State University, USA |
| Khokan Singha | M.Sc. Exploring the impact of Δ baryons on the properties of neutron stars |

Shahebaj Hasan Shaikh	M.Sc. Finite-temperature-dependent equation of state for the nucleonic core of neutron stars
Souhardya Sen	M.Sc. Radial oscillations in neutron stars from unified equations of state
Mrityunjay Singha	M.Sc. Dark compact objects
Vikram Singh Khatri	M.Sc. Presence of dark matter in the core of a neutron star
Manisha	M.Sc. Computation of two-fluid dark matter (SIRE project)
Arijit Maiti	M.Sc. Structure of neutron stars through numerical analysis of the TOV equations (SIRE project)
Pritam Kumar Bishee	B.Sc./B.Tech. Treatment of quark matter in the inner core of a neutron star
Soumya Prakash Behera	B.Sc./B.Tech. Nuclear constraints on gravitational waves from deformed pulsars

TEACHING

PH 5202	Nuclear and Particle Physics — Postgraduate, Even semester Nuclear structure, nuclear forces, radioactive decay, nuclear reactions, fission and fusion, introductory nuclear astrophysics.
PH 5301	Astrophysics — Postgraduate elective, Odd semester Stellar evolution, compact objects, cosmology, multi-messenger astronomy.
PH 4014	Atomic and Molecular Physics — Upper undergraduate, Even semester Atomic structure, molecular bonding, spectroscopy.
PH 3101	Quantum Mechanics — UG/PG, Odd semester The Schrödinger equation, operator formalism, the hydrogen atom, scattering theory.
PH 2102	Waves and Optics — Undergraduate, Even semester Wave mechanics, interference, diffraction, optical instruments; taught to engineering students across all disciplines.

INVITED AND CONTRIBUTED PRESENTATIONS

Invited Talks

1. Constraints on the moment of inertia of neutron stars/pulsars from GW170817, Physical Research Laboratory, Ahmedabad, India, 27 February 2020.
2. Relativistic models (G₃ and IOPB-I) of the neutron-star matter equation of state, Kyungpook National University, Daegu, South Korea, 24 September 2019.
3. Neutron star astrophysics from gravitational waves and nuclear theory, Sado2019 — 5th Workshop on Many-Body Correlations in Microscopic Nuclear Models, Senkaku-sou, Niigata, Japan, 5–6 September 2019.
4. Implications of nuclear interaction for nuclear structure and astrophysics within the relativistic mean-field model, Center for Computational Sciences, University of Tsukuba, Japan, 2019.

5. New relativistic effective interaction for finite nuclei, infinite nuclear matter and neutron stars, Tsukuba-CCS Workshop on “Microscopic Theories of Nuclear Structure and Dynamics”, University of Tsukuba, Japan.
6. New relativistic effective interaction for finite nuclei, infinite nuclear matter and neutron stars, International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru, India.

Contributed Talks and Posters

1. New effective interactions IOPB-I and G_3 , 18th CNS International Summer School, University of Tokyo, Japan, 2019.
2. Tidal deformability of neutron stars within relativistic mean field equations of state, CNT Lectures on Special Topics in Nuclear Astrophysics, Variable Energy Cyclotron Centre, Kolkata, India, 1–11 March 2017.
3. Tidal effects in equal-mass binary neutron stars, 61st DAE-BRNS Symposium on Nuclear Physics, Saha Institute of Nuclear Physics, Kolkata, India, 5–9 December 2016.
4. Examining the stability of thermally fissile Th and U isotopes, 40th Academic Year Activities of the Institute of Physics — IOP Internal Symposium, Bhubaneswar, India, 27 August 2015.
5. Shape coexistence and parity doublet in Zr isotopes, Nuclear Physics Meet 2015, Institute of Physics, Bhubaneswar, India, 26–30 June 2015.
6. Analysis of parity doublet in medium mass nuclei, DAE Symposium on Nuclear Physics, Department of Physics, Banaras Hindu University, India, 8–12 December 2014.

SCHOOLS, WORKSHOPS AND ACADEMIC VISITS

1. Quarks and Compact Stars 2019, Haeundae, Busan, South Korea, 26–28 September 2019.
2. Academic visit, Department of Physics Education, Daegu University, South Korea, 22–25 September 2019.
3. 18th CNS International Summer School, Hongo Campus, University of Tokyo, Japan, 21–27 August 2019.
4. 15th International Symposium on Origin of Matter and Evolution of Galaxies (OMEG15), Yukawa Institute for Theoretical Physics, Kyoto University, Japan, 2–5 July 2019.
5. India–UK Entrepreneurial Workshop, IUCAA, Pune, India, 18–19 January 2019.
6. Conference on Multi-messenger Astronomy in the Era of LIGO-India, Khandala, India, 15–18 January 2019.
7. 30th Meeting of the Indian Association for General Relativity and Gravitation (IAGRG), BITS-Pilani Hyderabad Campus, India, 3–5 January 2019.
8. Summer School on Gravitational-Wave Astronomy, International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru, India, 13–24 August 2018.
9. Summer School on Gravitational-Wave Astronomy, ICTS-TIFR, Bengaluru, India, 17–28 July 2017.
10. CNT Lectures on Special Topics in Nuclear Astrophysics, Variable Energy Cyclotron Centre, Kolkata, India, 1–11 March 2017.
11. Probing the Frontiers of Particle Physics with Neutrinos and the LHC, Institute of Physics, Bhubaneswar, India, 17–21 October 2016.

12. CNT Lectures on Selected Topics in Nuclear Theory, Variable Energy Cyclotron Centre, Kolkata, India, 16–25 February 2016.
13. International Workshop on Recent Trends in Nuclear Structure and its Implications in Astrophysics, Puri, India, 4–8 January 2016.
14. 40th Academic Year Activities of the Institute of Physics — IOP Internal Symposium, Bhubaneswar, India, 27 August 2015.
15. Recent Development of Density Functional Theory, Institute of Physics, Bhubaneswar, India, 17–19 July 2015.
16. Nuclear Physics Meet, Institute of Physics, Bhubaneswar, India, 26–30 June 2015.
17. Winter School on Nuclear Astrophysics, VECC, Kolkata, India, 19–31 January 2015.
18. DAE Symposium on Nuclear Physics, Department of Physics, Banaras Hindu University, India, 8–12 December 2014.
19. DST-SERC School on Nuclear Structure at High Angular Momentum and Isospin, HBCSE, TIFR, Mumbai, India, 5–25 October 2014.

SERVICE, AWARDS AND PROFESSIONAL STANDING

Peer Review

Since 2018	Physics Letters B • Modern Physics Letters A
Since 2019	The Astrophysical Journal • The Astrophysical Journal Letters
Since 2020	Physics of the Dark Universe
Since 2021	Nuclear Physics A

Institutional Service — NIT Rourkela

2022 – Present	Professor-in-Charge, Time Table and Departmental Seminars
2020 – Present	Member, Departmental Academic Committee (UG & PG)
2021 – 2022	Professor-in-Charge, Placement
2023	Member, XX Convocation Dress Distribution Committee

Professional Memberships

2022 – 2025	Associate, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India
Since 2021	Odisha Physical Society, India

Awards, Fellowships and Qualifications

2014 – 2018	Senior Research Fellowship (SRF), Department of Atomic Energy, India
2013 – 2014	Junior Research Fellowship (JRF), Department of Atomic Energy, India
2013	Qualified BARC Training School Examination (OCES & DGFS), Department of Atomic Energy, India
2013	Qualified Joint Entrance Screening Test (JEST), India
2011 – 2012	Post-Graduate Merit Scholarship, Aligarh Muslim University, India

REFERENCES

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